Lemma 3.6. *Let U be a symmetric and positive definite matrix. Let $y \in W^{2,2}(S;\mathbb{R}^3)$ be such that $\nabla y^T \nabla y = (U^2)'$. Then there exists a unique function $\nu \in W^{1,2}(S;\mathbb{R}^3)$ such that*

$$(\nabla y, \nu)U^{-1} \in \mathrm{SO}(3) \quad \textit{a.e. in } S\,.$$

In particular, ν is given by

$$\nu = \frac{1}{|U^{-1}e_3|^2}\left[\det(U^{-1})(\partial_1 y \wedge \partial_2 y) - \sum_{k=1}^{2}(U^{-1}e_k \cdot U^{-1}e_3)\partial_k y\right].$$

Proof. For the existence, it is enough to prove that $(\nabla y, \nu)^T(\nabla y, \nu) = U^2$ and $\det(\nabla y, \nu) > 0$. By the hypothesis on y we need to prove that

(i) $\partial_1 y \cdot \nu = (U^2)_{13}$,

(ii) $\partial_2 y \cdot \nu = (U^2)_{23}$,

(iii) $\nu \cdot \nu = (U^2)_{33}$.

For $j = 1, 2$ we have

$$\partial_j y \cdot \nu = -\frac{1}{|U^{-1}e_3|^2}\sum_{k=1}^{3}(U^{-1}e_k \cdot U^{-1}e_3)(U^2)_{jk} + (U^2)_{j3} = (U^2)_{j3}\,.$$

To complete the proof we observe that

$$|Ue_1 \wedge Ue_2|^2 = |Ue_1|^2|Ue_2|^2 - (Ue_1 \cdot Ue_2)^2 = |\partial_1 y \wedge \partial_2 y|^2$$

and, since $Ue_1 \wedge Ue_2 = \mathrm{cof}(U)e_3$

$$|Ue_1 \wedge Ue_2|^2 = |\det(U)U^{-1}e_3|^2 = \det^2(U)|U^{-1}e_3|^2\,.$$

We are now ready to conclude:

$$\begin{aligned}
\nu \cdot \nu &= \frac{1}{|U^{-1}e_3|^2} + \frac{(U^{-1}e_1 \cdot U^{-1}e_3)}{|U^{-1}e_3|^4}\sum_{k=1}^{3}(U^{-1}e_k \cdot U^{-1}e_3)(U^2)_{1k} \\
&\quad + \frac{(U^{-1}e_2 \cdot U^{-1}e_3)}{|U^{-1}e_3|^4}\sum_{k=1}^{3}(U^{-1}e_k \cdot U^{-1}e_3)(U^2)_{2k} \\
&\quad - \frac{1}{|U^{-1}e_3|^2}\sum_{k=1}^{3}(U^{-1}e_k \cdot U^{-1}e_3)(U^2)_{3k} + (U^2)_{33} \\
&= \frac{1}{|U^{-1}e_3|^2} - \frac{1}{|U^{-1}e_3|^2} + (U^2)_{33} = (U^2)_{33}\,.
\end{aligned}$$

To show that $\det(\nabla y, \nu) > 0$, it is sufficient to note that

$$(\nabla y, \nu) = (\partial_1 y, \partial_2 y, \partial_1 y \wedge \partial_2 y) \cdot \begin{pmatrix} 1 & 0 & -\frac{1}{|U^{-1}e_3|^2}(U^{-1}e_1 \cdot U^{-1}e_3) \\ 0 & 1 & -\frac{1}{|U^{-1}e_3|^2}(U^{-1}e_2 \cdot U^{-1}e_3) \\ 0 & 0 & \frac{\det(U^{-1})}{|U^{-1}e_3|^2} \end{pmatrix},$$

and that the determinant of both matrices in the right-hand side is positive.

To prove uniqueness we observe that, for any choice of two different rotations $R_1, R_2 \in \mathrm{SO}(3)$, we have $\mathrm{rank}(R_1 - R_2) \neq 1$. Indeed, given a vector $v \in \mathbb{R}^3$ we have

$$(R_1 - R_2)v = 0 \iff R_1 v = R_2 v \iff R_2^T R_1 v = v\,.$$

Since $R_2^T R_1 \in \mathrm{SO}(3)$ and $R_2^T R_1 \neq \mathrm{Id}$, we deduce that $v \in \ker(R_1 - R_2)$ if and only if v is parallel to the rotation axis, that is, v belongs to a 1-dimensional subspace. In particular, $\mathrm{rank}(R_1 - R_2) = 2$. Suppose now that there is another vector $\tilde{\nu}$ such that

$$(\nabla y, \tilde{\nu})U^{-1} \in \mathrm{SO}(3) \text{ a.e..}$$

Then,

$$(0, \nu - \tilde{\nu})U^{-1} = (\nu - \tilde{\nu}) \otimes U^{-1} e_3$$

coincides a.e. with the difference of two rotations and has rank 1 whenever $\nu \neq \tilde{\nu}$. Thus, $\nu = \tilde{\nu}$ almost everywhere. ■

We move now to the proof of the first part of Theorem 1.1.

Proof of Theorem 1.1–(i). By Corollary 3.3 there is a sequence of indices $i_h \in \{1, \dots, l\}$ such that

$$\| \operatorname{dist}(\nabla_h y_h, K_{i_h})\|_{L^2} \leq Ch\,.$$

Upon a further subsequence, since i_h takes values in a finite set, we can suppose i_h to be constant and equal to j. Construct the sequences R_h and $\tilde{R}_h$ as in Proposition 3.4. Clearly, $\tilde{R}_h$ is bounded in $W^{1,2}(S; \mathbb{R}^{3\times 3})$ thus it converges weakly, at least along a subsequence, to a map $R \in W^{1,2}(S; \mathbb{R}^{3\times 3})$. Hence, we have $R_h \to R$ in $L^2(S; \mathrm{SO}(3))$, so R takes values in the set of rotations. Consequently, $\nabla_h y_h U_j^{-1} \to R$ in $L^2(S; \mathbb{R}^{3\times 3})$. By an application of the Poincaré-Wirtinger inequality we have

$$\|y_h - c_h\|_{W^{1,2}} \leq C\|\nabla_h y_h\|_{L^2} \leq C\,,$$

where

$$c_h = \frac{1}{|\Omega|}\int_\Omega y_h(x)\,dx\,.$$

Thus, $y_h - c_h$ converges weakly (possibly along a subsequence) to some map $y \in W^{1,2}(\Omega; \mathbb{R}^3)$. Since $\frac{1}{h}\partial_3 y_h$ is bounded in $L^2(\Omega; \mathbb{R}^3)$, we have $\partial_3 y = 0$ and $y \in W^{1,2}(S; \mathbb{R}^3)$. Hence, $\nabla y = (RU_j)^{1,2}$, $y \in W^{2,2}(S; \mathbb{R}^3)$ and $\nabla'_h y_h \to \nabla y$. Lastly, since ν is uniquely determined by the condition $(\nabla y, \nu)U_j^{-1} \in \mathrm{SO}(3)$ a.e., the remaining part follows from Lemma 3.6. ■

Corollary 3.7. *In the same setting of Theorem 1.1, there is a sequence $(R_h) \subset L^\infty(S; \mathrm{SO}(3))$ such that, up to a subsequence*

$$G_h := h^{-1}\left(R_h^T \nabla_h y_h - U_j\right) \rightharpoonup G \text{ in } L^2(\Omega; \mathbb{R}^{3\times 3})\,. \tag{17}$$

Moreover, $G^{1,2}$ is affine in x_3, that is

$$G^{1,2}(x', x_3) = G_0(x') + x_3 G_1(x')\,.$$

Lastly,

$$(U_j G_1)' = \nabla y^T \nabla \nu\,. \tag{18}$$

Proof. Arguing as in the proof of Theorem 1.1–(i) we have

$$\| \operatorname{dist}(\nabla_h y_h, K_j)\|_{L^2} \leq Ch\,,$$

for some $j \in \{1, \dots, l\}$. Let $R_h \in L^\infty(S; \mathrm{SO}(3))$ be the map given by Proposition 3.4. Convergence (17) follows from (R1). Moreover, arguing again as in the proof of Theorem 1.1–(i) we deduce that, up to subsequences, $R_h \to R \in L^2(S; \mathrm{SO}(3))$ and $\nabla_h y_h U_j^{-1} \to (\nabla y, \nu) U_j^{-1} = R$ in $L^2(\Omega; \mathbb{R}^{2\times 2})$, where ν is given by Lemma 3.6. Define

$$H_h^s(x', x_3) = \frac{1}{s}(G_h(x', x_3 + s) - G_h(x', x_3))\,,$$

for s such that $x_3 + s \in I$. For $\alpha = 1, 2, 3$ and $\beta = 1, 2$ we have

$$\begin{aligned}(R_h(x')H_h^s(x', x_3))_{\alpha\beta} &= \frac{1}{s}h^{-1}\left(\frac{\partial y_{h,\alpha}}{\partial x_\beta}(x', x_3 + s) - \frac{\partial y_{h,\alpha}}{\partial x_\beta}(x', x_3)\right) = \\ &= \frac{1}{s}\frac{\partial}{\partial x_\beta}\int_0^s \frac{1}{h}\frac{\partial y_{h,\alpha}}{\partial x_3}(x', x_3 + \sigma)\, d\sigma\,.\end{aligned}$$

The right-hand side converges strongly in $(W_0^{1,2}(\Omega))^*$ to $(\nabla\nu)_{\alpha\beta}$ as $h \to 0$. Indeed, one has that $\|\partial_i g\|_{(W_0^{1,2})^*} \leq \|g\|_{L^2}$ for every $g \in L^2(\Omega)$, where $\|\cdot\|_{(W_0^{1,2})^*}$ is the standard operatorial norm. The left-hand side converges weakly in $L^2(\Omega)$ to

$$((\nabla y, \nu)U_j^{-1}H^s(x', x_3))_{\alpha\beta}\,,$$

where

$$H^s(x', x_3)_{\alpha\beta} = \frac{1}{s}(G(x', x_3 + s)_{\alpha\beta} - G(x', x_3)_{\alpha\beta})\,.$$

Since $L^2(\Omega)$ is continuously embedded in $(W_0^{1,2}(\Omega))^*$ we obtain

$$H^s(x', x_3)_{\alpha\beta} = (U_j^{-1}(\nabla y, \nu)^T\nabla\nu)_{\alpha\beta} \quad \forall\, \alpha = 1, 2, 3 \quad \forall\, \beta = 1, 2\,.$$

In particular, the first two columns of H^s are independent of x_3 and so the first two columns of G are affine in x_3. Finally, we have

$$(U_j G_1)' = ((\nabla y, \nu)^T\nabla\nu)_{1,2} = \nabla y^T\nabla\nu\,,$$

that proves (18). ∎

3.2 Compactness for the regime $\alpha > 2$

In the case $\alpha > 2$ we will write the Γ-limit as a function of the in-plane and out-of-plane displacements. The next results give the correct scaling to extract their convergence. We start with a preliminary lemma.

Lemma 3.8. *Let $(F_h) \subset \mathbb{R}^{3\times 3}$ be a sequence such that $|F_h - \mathrm{Id}\,| \leq Ch^\beta$ for some $\beta > 0$. Then there is a sequence $(P_h) \subset \mathrm{SO}(3)$ such that $\operatorname{skew}(P_h F_h) = 0$ and $|P_h - \mathrm{Id}\,| \leq Ch^\beta$.*

Proof. For $h \ll 1$ the matrix F_h is invertible with positive determinant and so its polar decomposition $F_h = R_h A_h$ is uniquely determined providing a matrix $P_h \in \mathrm{SO}(3)$ (i.e., $P_h = R_h^T$) such that $P_h F_h$ is symmetric. We have that

$$|F_h - R_h| \leq |F_h - X| \quad \forall\, X \in \mathrm{O}(3) = \{B \in \mathbb{R}^{3\times 3} \,:\, \det(B) = \pm 1\}\,. \tag{19}$$